

Advanced Programming 1

Assignment 3 – Project Design & Setup Milestone

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Group: SE 2438

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1. Project Proposal

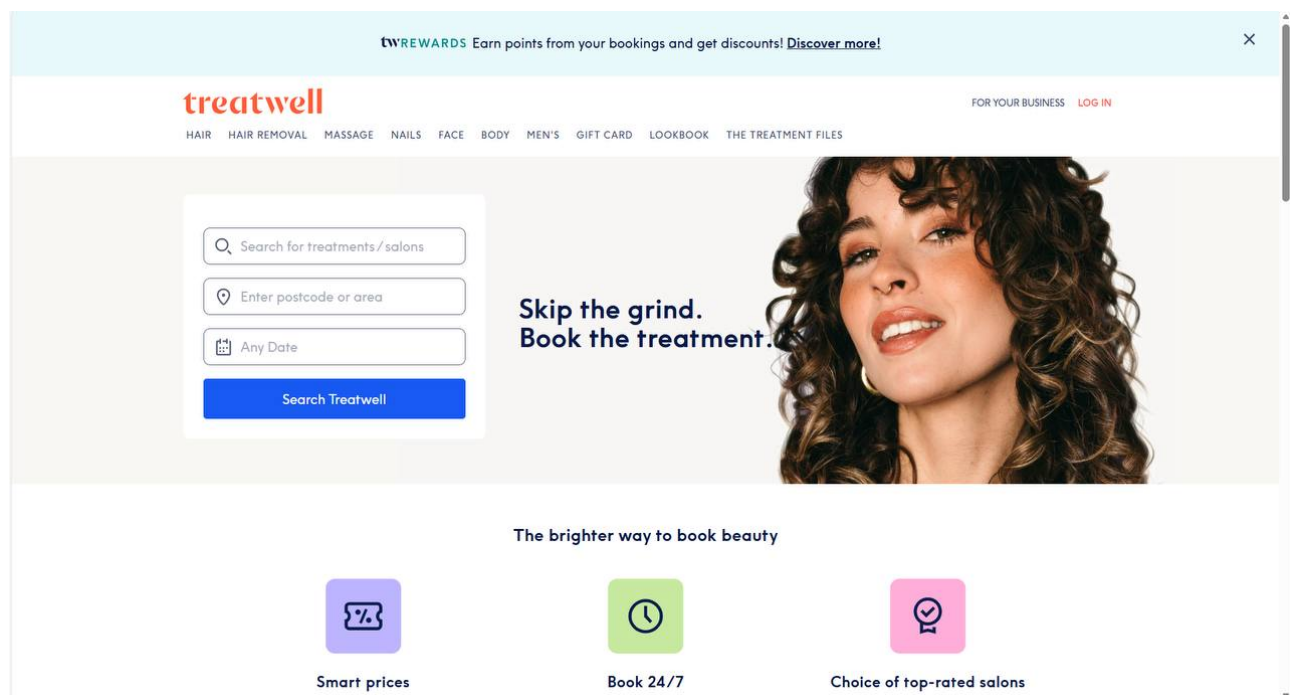
Project relevance

Many beauty salons in Kazakhstan still manage appointments through WhatsApp, Instagram, or phone calls. This manual process often causes missed messages, double bookings, and scheduling conflicts. Clients cannot quickly check free time slots, compare service prices and duration, or choose a specific master without extra communication. Administrators spend a lot of time confirming bookings manually and handling cancellations or rescheduling. A centralized online booking system solves these problems by providing clear scheduling, structured service information, and organized appointment management.

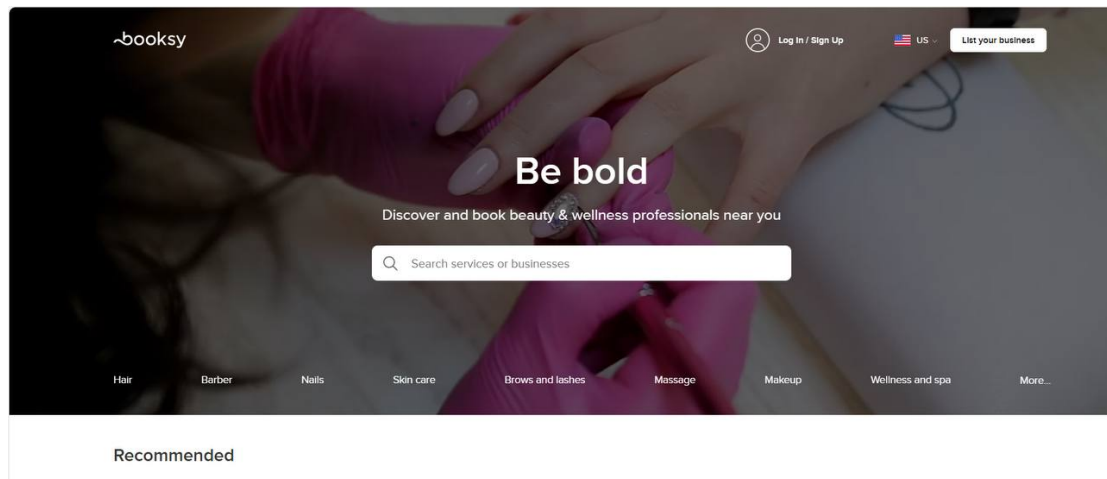
Competitor Analysis

We explored several popular services:

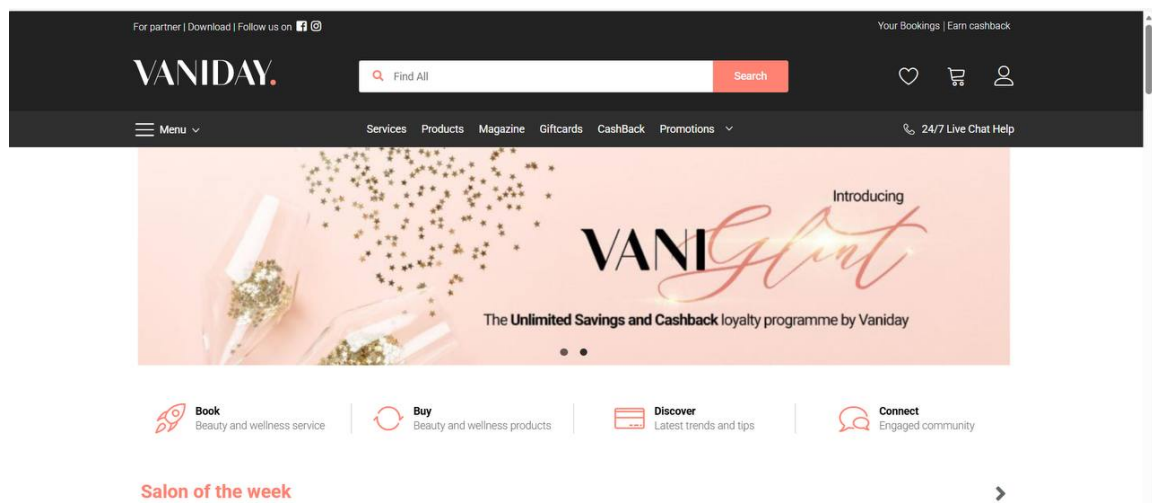
Treatwell - easy to book, but the interface is sometimes cluttered with ads.



Booksy – good mobile version, but limited features for salon administrators



Vaniday – simple booking, but little analytics for salon owners.



Our project will focus on convenience for clients and ease of management for the salon, as well as providing basic analytics on visits and popular services.

Target users

- Clients (Customers): book services online, choose a master, and manage their appointments.
- Salon Administrator: manages the schedule, confirms/cancels bookings, and controls services and masters.
- Masters (Employees): view their personal timetable and upcoming client appointments.

Planned features

- Online booking: choose service, master, date, and time
- Service catalog: services with price, duration, description
- User accounts: client profile + booking history
- Appointment management: cancel or reschedule appointments
- Notifications: booking confirmation + reminders
- Admin dashboard: manage bookings + view basic statistics

2. Architecture & Design

System Architecture:

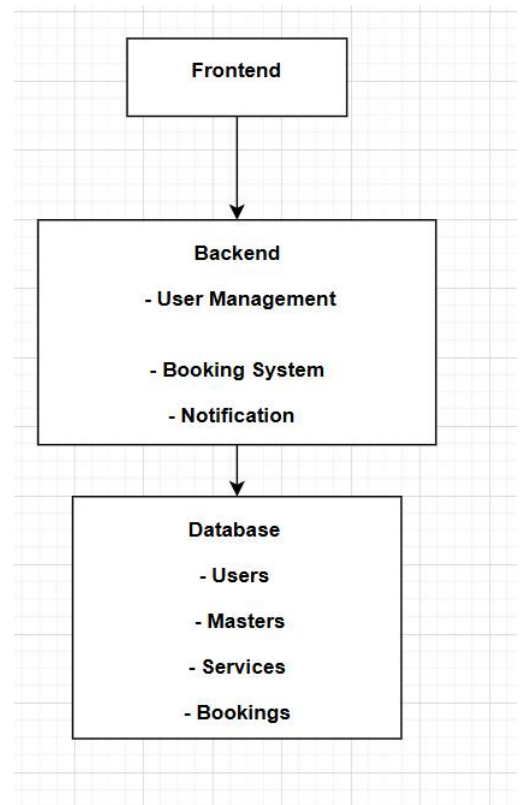
Module breakdown (monolith):

The backend is implemented as a monolithic Go REST API. It contains:

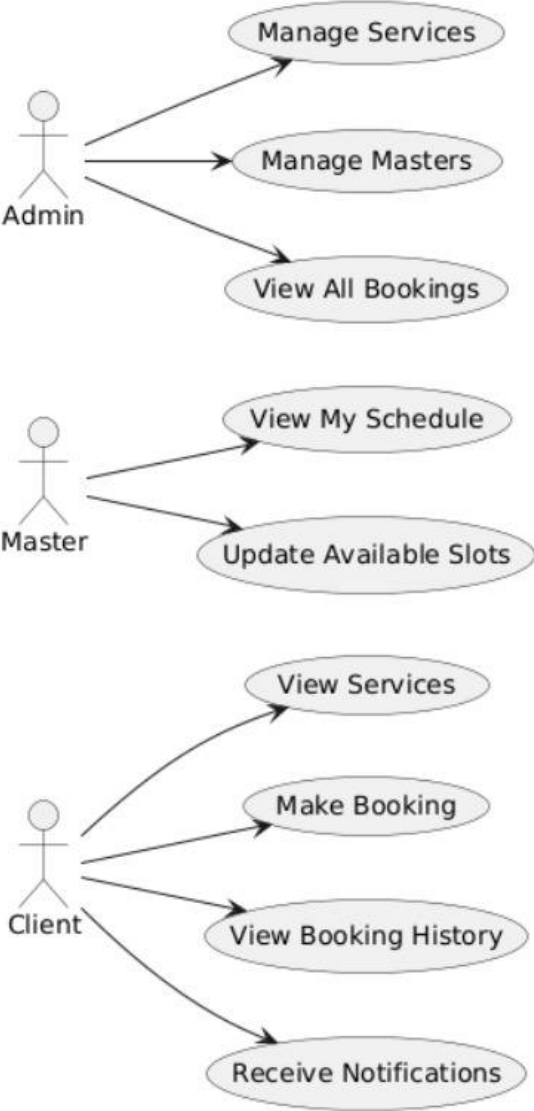
- Handlers (routes/controllers): receives HTTP requests, parses input, and returns JSON responses.
- Service layer (business logic): performs validation, booking rules (busy-check), and status updates (booked/cancelled).
- Repository (data access): handles CRUD operations and database queries for users, services, masters, appointments, and payments.
- Models: Go structs that represent database entities (Users, Masters, Services, MasterServices, Appointments, Payments).

Request flow example (Create booking):

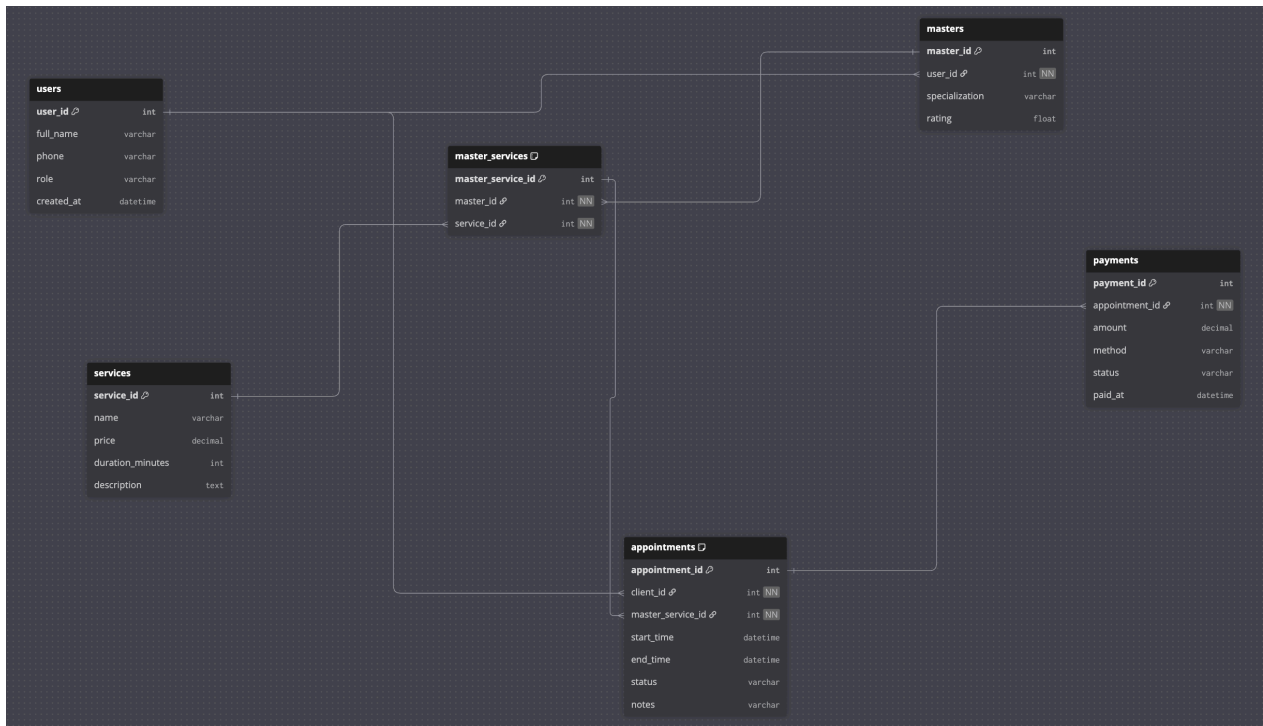
Client → POST /appointments → handler → booking service (validation + busy rule) → repository → database → JSON response (created appointment or error).



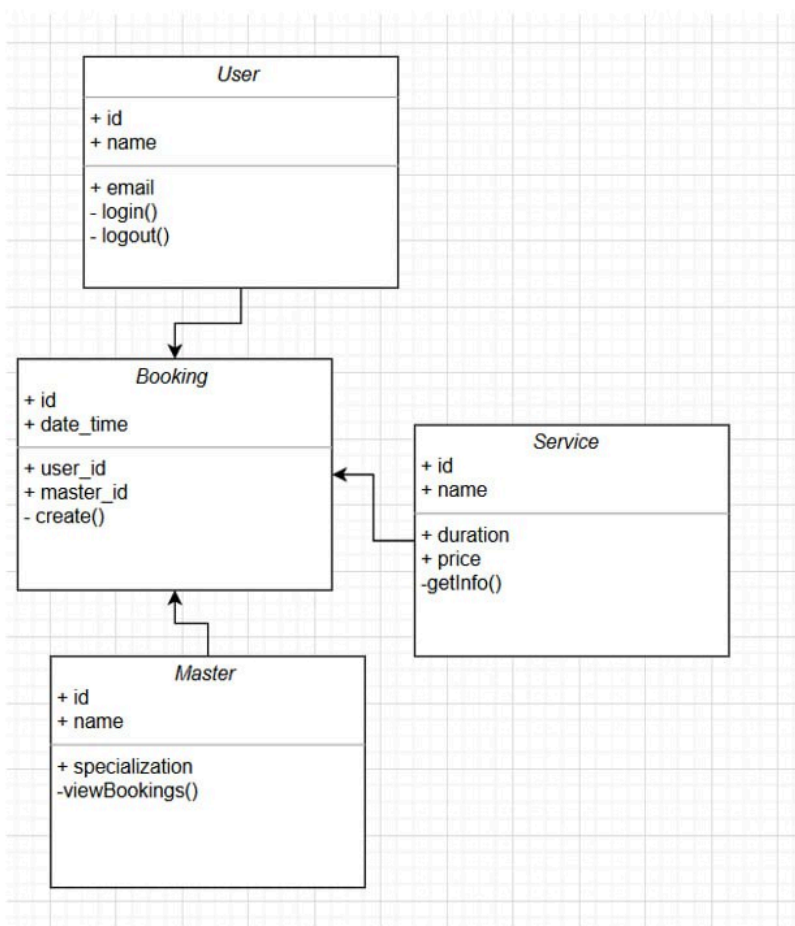
Use-Case diagram



ERD



UML diagram.



3. Project Plan

Week	Student A (Assem)	Student B (Nursaya)
Week 7 (Design + Setup)	Create ERD + database structure (Users, Masters, Services, Appointments). Create repo + base project structure (go run.). Define models and fields	Create Use-Case + UML (draft) . Write Architecture draft (modules + request flow). Review A's models and confirm fields/relationships
Week 8 (Core Backend MVP)	Implement CRUD Services + validation (required fields, price, duration). Prepare Service data and master-service mapping	Implement Booking create endpoint + busy rule + validation. Cross task: implement 1 Services endpoint (GET /services or search/filter)
Week 9 (Admin + Improvements)	Implement Admin features: add masters + connect Master ↔ Services. Cross task: implement 1 client endpoint (example: available time slots / filter slots)	Implement Client features: view services, masters, user appointments list. Improve UML/Use-Case + add usage examples to documentation
Week 10 (Demo + Final Ready)	Testing + fixes + final merge. Prepare demo scenario (create booking → cancel/reschedule). Presentation (A part): backend/admin/endpoints/validation/demo	Fix 1–2 bugs (error handling, edge cases, API responses) + commit. Final documentation formatting. Presentation (B part): booking/client/UML-UseCase/future improvements